# O'Neill §1.2 Problem Set

**Exercise 1.** *Let* $\mathbf{v} = (-2, 1, -1)$ *and* $\mathbf{w} = (0, 1, 3)$.

*(a) At an arbitrary point* $\mathbf{p}$, *express the tangent vector* $3\mathbf{v_p} - 2\mathbf{w_p}$ *as a linear combination of* $U_1(\mathbf{p}), U_2(\mathbf{p}), U_3(\mathbf{p})$.

*(b) For* $\mathbf{p} = (1, 1, 0)$, *make an accurate sketch showing the four tangent vectors* $\mathbf{v_p}$, $\mathbf{w_p}$, $-2\mathbf{v_p}$, *and* $\mathbf{v_p} + \mathbf{w_p}$.

*Solution.* (a) We compute the scalar components directly:

$$3\mathbf{v} - 2\mathbf{w} = 3(-2, 1, -1) - 2(0, 1, 3) = (-6, 3, -3) - (0, 2, 6) = (-6, 1, -9).$$

Expressing this vector at the point $\mathbf{p}$ in terms of the natural frame field yields:

$$3\mathbf{v_p} - 2\mathbf{w_p} = -6U_1(\mathbf{p}) + U_2(\mathbf{p}) - 9U_3(\mathbf{p}).$$

(b) *(Description for sketching)* At the origin point $\mathbf{p} = (1, 1, 0)$, draw the standard coordinate axes translated to this point. Then draw the following arrows rooted at $(1, 1, 0)$:

- $\mathbf{v_p}$ points to $(1 - 2, 1 + 1, 0 - 1) = (-1, 2, -1)$.
- $\mathbf{w_p}$ points to $(1 + 0, 1 + 1, 0 + 3) = (1, 2, 3)$.
- $-2\mathbf{v_p} = (4, -2, 2)_\mathbf{p}$ points to $(1 + 4, 1 - 2, 0 + 2) = (5, -1, 2)$.
- $\mathbf{v_p} + \mathbf{w_p} = (-2, 2, 2)_\mathbf{p}$ points to $(1 - 2, 1 + 2, 0 + 2) = (-1, 3, 2)$.

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**Exercise 2.** *Let* $V = xU_1 + yU_2$ *and* $W = 2x^2U_2 - U_3$. *Compute the vector field* $W - xV$, *and find its value at the point* $\mathbf{p} = (-1, 0, 2)$.

*Solution.* First, find the expression for the vector field $W - xV$:

$$\begin{aligned} W - xV &= (2x^2U_2 - U_3) - x(xU_1 + yU_2) \\ &= -x^2U_1 + (2x^2 - xy)U_2 - U_3. \end{aligned}$$

Now evaluate this vector field at the point $\mathbf{p} = (-1, 0, 2)$ where $x = -1, y = 0, z = 2$:

$$\begin{aligned} (W - xV)(\mathbf{p}) &= -(-1)^2U_1(\mathbf{p}) + (2(-1)^2 - (-1)(0))U_2(\mathbf{p}) - U_3(\mathbf{p}) \\ &= -U_1(\mathbf{p}) + 2U_2(\mathbf{p}) - U_3(\mathbf{p}) = (-1, 2, -1)_\mathbf{p}. \end{aligned}$$

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**Exercise 3.** *In each case, express the given vector field $V$ in the standard form $\sum v_i U_i$.*

*(a)* $2z^2U_1 = 7V + xyU_3$.

*(b)* $V(\mathbf{p}) = (p_1, p_3 - p_1, 0)_{\mathbf{p}}$ *for all* $\mathbf{p}$.

*(c)* $V = 2(xU_1 + yU_2) - x(U_1 - y^2U_3)$.

*(d) At each point* $\mathbf{p}$, $V(\mathbf{p})$ *is the vector from the point* $(p_1, p_2, p_3)$ *to the point* $(1 + p_1, p_2p_3, p_2)$.

*(e) At each point* $\mathbf{p}$, $V(\mathbf{p})$ *is the vector from* $\mathbf{p}$ *to the origin.*

*Solution.* (a) Rearranging the equation to solve for $V$:

$$7V = 2z^2U_1 - xyU_3 \implies V = \frac{2z^2}{7}U_1 - \frac{xy}{7}U_3.$$

(b) Using standard coordinate variables $x = p_1$, $y = p_2$, $z = p_3$:

$$V = xU_1 + (z - x)U_2 + 0U_3 = xU_1 + (z - x)U_2.$$

(c) Expanding and grouping terms by basis element:

$$V = 2xU_1 + 2yU_2 - xU_1 + xy^2U_3 = xU_1 + 2yU_2 + xy^2U_3.$$

(d) The components are computed by subtracting the components of the initial point from the terminal point:

$$v_1 = (1 + p_1) - p_1 = 1, \quad v_2 = p_2p_3 - p_2, \quad v_3 = p_2 - p_3.$$

Substituting the variables $x, y, z$ gives:

$$V = U_1 + (yz - y)U_2 + (y - z)U_3.$$

(e) The vector from $\mathbf{p} = (x, y, z)$ to the origin $(0, 0, 0)$ is $(0-x, 0-y, 0-z) = (-x, -y, -z)$:

$$V = -xU_1 - yU_2 - zU_3.$$

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**Exercise 4.** *If* $V = y^2U_1 - x^2U_3$ *and* $W = x^2U_1 - zU_2$, *find functions* $f$ *and* $g$ *such that the vector field* $fV + gW$ *can be expressed in terms of* $U_2$ *and* $U_3$ *only.*

*Solution.* We write out the linear combination $fV + gW$:

$$fV + gW = f(y^2 U_1 - x^2 U_3) + g(x^2 U_1 - zU_2) = (fy^2 + gx^2)U_1 - gzU_2 - fx^2 U_3.$$

For this field to be expressed in terms of $U_2$ and $U_3$ only, the $U_1$ coefficient component must vanish identically everywhere:

$$fy^2 + gx^2 = 0 \implies fy^2 = -gx^2.$$

A simple, non-trivial solution choice satisfies this criteria by switching coefficients:

$$f = x^2, \quad g = -y^2.$$

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**Exercise 5.** *Let $V_1 = U_1 - xU_3$, $V_2 = U_2$, and $V_3 = xU_1 + U_3$.*

*(a) Prove that the vectors $V_1(\mathbf{p}), V_2(\mathbf{p}), V_3(\mathbf{p})$ are linearly independent at each point of $\mathbf{E}^3$.*

*(b) Express the vector field $xU_1 + yU_2 + zU_3$ as a linear combination of $V_1, V_2, V_3$.*

*Solution.* (a) We set up the coefficient components as column vectors in a matrix to check the determinant:

$$\det \begin{pmatrix} 1 & 0 & x \\ 0 & 1 & 0 \\ -x & 0 & 1 \end{pmatrix} = 1(1-0) - 0 + x(0 - (-x)) = 1 + x^2.$$

Since $x^2 \geq 0$ for all real numbers, $1 + x^2 \geq 1 > 0$ everywhere. Because the determinant is never zero, the fields remain linearly independent at every point $\mathbf{p} \in \mathbf{E}^3$.

(b) We seek functions $a, b, c$ such that $aV_1 + bV_2 + cV_3 = xU_1 + yU_2 + zU_3$. Expanding:

$$a(U_1 - xU_3) + b(U_2) + c(xU_1 + U_3) = (a + cx)U_1 + bU_2 + (-ax + c)U_3.$$

Equating corresponding components yields a system of equations:

$$1)\ a + cx = x, \quad 2)\ b = y, \quad 3)\ -ax + c = z.$$

From (3), $c = z + ax$. Substitute this into (1):

$$a + (z + ax)x = x \implies a(1 + x^2) + xz = x \implies a = \frac{x - xz}{1 + x^2}.$$

Now back-substitute to find $c$:

$$c = z + x\left(\frac{x - xz}{1 + x^2}\right) = \frac{z(1 + x^2) + x^2 - x^2 z}{1 + x^2} = \frac{z + x^2}{1 + x^2}.$$

Thus, the linear combination expression is:

$$\left(\frac{x - xz}{1 + x^2}\right)V_1 + yV_2 + \left(\frac{z + x^2}{1 + x^2}\right)V_3.$$

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